AN M-REGULAR SEQUENCE IS M-KOSZUL-REGULAR

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Koszul complex of a sequence relative to a module and its homology, and the definition of M-Koszul-regularity. used throughout the proof, lines 7273-7298. Supplies the goal and the complex through which it is tested.
- The Koszul complex of a longer sequence is the cone on multiplication by the first element on the Koszul complex of the tail. the author cites Lemma lemma-cone-koszul (tag 0629) at proof lines 7281-7284. Is the structural statement that converts the problem into a statement about the cone.
- The long exact sequence of cohomology attached to a cone and the acyclicity criterion for a cone. used at proof lines 7284-7297. Transfers the vanishing of the homology from the tail to the whole sequence.

Source reference. Original lemma 062F: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7794–7794. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0629: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7788–7788. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring, M an R-module, and $f_1, \ldots, f_r \in R$ such that for $i = 1, \ldots, r$ multiplication by f_i is injective on $M/(f_1, \ldots, f_{i-1})M$. Then $f_1, \ldots, f_r$ is M-Koszul regular. In particular, an M-regular sequence is M-Koszul-regular and any regular sequence is Koszul-regular.*

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Proof. Let R, M, $f_1, \ldots, f_r$ be as in the first sentence of the lemma. If $r = 1$, it is immediate that f_1 is M-Koszul-regular. Assume $r > 1$. Since f_1 is a nonzerodivisor on M, we obtain a short exact sequence of complexes:

$$0 \to K_\bullet(f_2, \ldots, f_r) \otimes M \xrightarrow{f_1} K_\bullet(f_2, \ldots, f_r) \otimes M \to K_\bullet(\overline{f}_2, \ldots, \overline{f}_r) \otimes M/f_1M \to 0$$

Here $\overline{f}_i$ is the image of f_i in $R/(f_1)$. By Lemma 0629 the complex $K_\bullet(f_1, \ldots, f_r)$ is isomorphic to the cone of multiplication by f_1 on $K_\bullet(f_2, \ldots, f_r)$. Thus $K_\bullet(R, f_1, \ldots, f_r) \otimes M$ is isomorphic to the cone on the first map. Hence $K_\bullet(\overline{f}_2, \ldots, \overline{f}_r) \otimes M/f_1M$ is quasi-isomorphic to $K_\bullet(f_1, \ldots, f_r) \otimes M$. As $R/(f_1)$, M/f_1M, $\overline{f}_2, \ldots, \overline{f}_r$ satisfy the conditions of the lemma, by induction we conclude this complex is acyclic in postive degrees. This finishes the proof of the first statement. The second statement immediately follows from the first. $\square$